

Treatment allocation under spatial spillover: heterogeneous incidence and education-pilot planning

Andrew Walther¹, Ross Joseph Simpson, Jr.², Ashkan Habib¹, and Feng-Chang Lin¹

¹Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA

²Division of Cardiology, University of North Carolina School of Medicine, Chapel Hill, NC, USA

Correspondence: Feng-Chang Lin, Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599, USA. Email: flin33@email.unc.edu. Telephone: [AUTHOR TO SUPPLY].

Abstract

Background/Aims: Spatial spillover can make treatment-effect estimation depend on how clusters are assigned to treatment. We compared average estimation error and allocation-specific downside across eight assignment strategies and simple random sampling (SRS), including a completed application for planning a sudden unexpected death (SUD) prevention education pilot.

Methods: A 100-cluster grid study varied incidence heterogeneity, spatial dependence, spillover magnitude, spillover regime and treatment-effect size. Maximum likelihood spatial autoregressive models estimated a known structural intervention effect. A focused nested pilot estimated conditional MSE across allocations. The application held observed 2018–2021 SUD incidence fixed for 58 North Carolina community-college service-area clusters and simulated continuous education outcomes. Primary application incidence guided assignment without entering the outcome baseline. Queen weights were primary; rook, hypothetical baseline/adjustment and regional-summary sensitivities were assessed.

Results: On the grid at effect size one, incidence-guided saturation had control-only mean MSE 0.1355 versus SRS 0.2146, while SRS remained competitive under both-arms spillover. In the completed application, control-only mean MSE was 27.4% lower for plain saturation and 21.1–29.9% lower for guided saturation than SRS, depending on year. Refined allocation-tail estimates also favored saturation under control-only spillover, with uncertain worst-decile membership. Under both arms, SRS and several randomized strategies had similar mean accuracy. Application yearly mean coverage was approximately 92.6–93.6%, below nominal 95%. All prespecified application Monte Carlo precision gates passed, with no warnings, aliases, boundary fits or estimation failures.

Conclusions: Saturation merits consideration when spillover primarily reaches controls; SRS remains competitive when both arms receive spillover. Incidence guidance did not establish an additional accuracy benefit over plain saturation. Recommendations require attention to model assumptions, treatment budgets, interval undercoverage and finite allocation-tail precision. The application evaluates trial design, not observed intervention effectiveness.

Keywords: cluster randomized trial; spatial spillover; treatment allocation; simulation; mean squared error; sudden unexpected death.

Reading preview only. Ethics/data-use documentation and final author approval remain pending.

Introduction

Cluster randomized trials assign groups rather than individuals when interventions operate through clinics, schools or geographic areas.^{1,2} Spatial interference challenges the assumption that a cluster's outcome depends only on its own treatment: adjacent populations can share services, information or environmental exposures.^{3,4} An allocation may therefore provide substantially different information about a treatment effect than another allocation with the same number of treated clusters.

Design-stage methods include covariate-constrained randomization, randomized saturation and independent-set or spatial cluster designs.^{5–8} They address different objectives and constraints. Balancing measured incidence does not ensure treatment and spillover are distinguishable; maximal interspersed can itself make those covariates dependent. Appropriate spatial analysis remains necessary alongside allocation design.

In previous work, we compared simple random assignment and block-stratified sampling on small grids.⁹ Its practical argument considered reasonably good estimation together with limiting poor-allocation downside in the studied settings. We extend that question to heterogeneous incidence, a larger grid, additional strategies and an irregular observed geography. We include both SRS and the strict alternating allocation called Checkerboard, while evaluating mean MSE and the distribution of conditional MSE across assignments. We do not directly compare numerical results between studies.

Our application supports planning an education intervention for healthcare professionals involved in SUD prevention. Historical incidence can identify areas of need without implying that education ability, or the intervention's effect on it, depends on incidence. We consequently separate incidence-guided allocation from the simulated education baseline. A trial may realize only one assignment, making both expected accuracy and poor-allocation downside relevant.

Methods

Grid model and assignments

We used a 10×10 grid with row-standardized weights W . Queen neighbors share an edge or corner; rook neighbors share an edge. Queen was primary, with rook assessed separately. Outcomes followed

$$Y = (I - \rho W)^{-1}[\tau Z + S(Z) + \beta X + \epsilon], \quad (1)$$

$$\epsilon \sim N(0, I),$$

where Z is treatment and τ its structural coefficient before spatial propagation. It is not an average total effect after applying the spatial multiplier. We set $\beta = 1$ in the grid study. Spillover was $S(Z) = \gamma W Z$ for both arms or $S(Z) = \gamma(1 - Z) \circ W Z$ for controls only, with elementwise product $\circ$. The parameter grid crossed $\rho = 0, 0.01, 0.2, 0.5$, $\gamma = 0.5, 0.6, 0.7, 0.8$ and $\tau = 0.8, 1, 1.5, 2, 3$.

Five incidence configurations comprised independent Uniform(0,1) incidence, spatial incidence at two dependence strengths and Poisson incidence at the same two

strengths. Spatial incidence used a queen SAR filter followed by a normal distribution-function transformation. Poisson counts used spatial log-rates with base rate 35 per 100,000 and equal populations of 100,000; observed rates were rank-scaled to $(r_i - 0.5)/100$, using average ranks for ties. These are general simulation settings, not calibration to the application counts. Each configuration's ten independent surfaces were shared across parameters. For surface k , assignment, outcome generation and fitted incidence all used X_k .

The eight strategies were: Checkerboard; treatment of the highest-incidence half; plain saturation quadrants; Isolation Buffer, a greedy random maximal independent set; randomization of two clusters in every 2×2 block; balancing by incidence quartiles; balancing by incidence halves; and incidence-guided saturation quadrants. Both saturation strategies assigned 20/40/60/80% treatment within four quadrants; plain saturation randomized their order and guided saturation ordered them by mean incidence. All treated 50 clusters except buffer, whose budget depended on the graph. SRS completely randomized 50 of 100. Supplementary material gives exact rules. Checkerboard was deterministic; incidence ties were broken randomly for each assignment in ranking-based designs. High Incidence Focus had singleton support when no tie affected selection.

The nine-design benchmark contained 14,400 scenarios per estimator, each with ten surfaces and 25 assignment/outcome pairs per surface, giving 250 independently noised fits. Designs within a block shared surfaces and outcome noise; treatment-effect levels reused assignment/noise draws. The SRS extension preserved the original eight-design outputs. Scenario metrics were MSE, bias and 95% Wald coverage. Surface-level Monte Carlo SEs and paired comparisons used independent configuration-by-surface units rather than treating shared-surface blocks as independent replications. Detailed six-design ranking analyses remain explicitly subset analyses in the supplement.

Estimation and allocation risk

The primary maximum likelihood SAR fit included an intercept, treatment, the true regime-specific spillover covariate and grid incidence. Our concentrated-likelihood implementation matched `spatialreg::lagsarlm` in validation; confidence intervals were $\hat{\tau} \pm 1.96\text{SE}$.^{10,11} This assumes the analyst knows the spillover form. A grid sensitivity omitted spillover. Aliases, warnings, failed estimates/intervals and matrix rank were recorded. Rook Checkerboard has $WZ = 1 - Z$, so treatment and spillover are not separately identified; its retained treatment coefficient estimates $\tau - \gamma$. Under queen control-only spillover, identification relies heavily on the grid boundary. The rook estimate does not recover the target effect; queen Checkerboard remains identified but can be imprecise.

For assignment a , allocation risk is $m(a) = E_\epsilon[(\hat{\tau} - \tau)^2 | a, X]$. Nested outcome replication estimates its distribution before calculating allocation q90 and the estimated mean of the worst 10% of sampled assignments. These differ from quantiles of realized errors or scenario-level MSE. A focused grid pilot used two fixed surfaces per configuration, queen parameter corners, 100 assignment draws and 100 outcomes per unique assignment, with a selected same-assignment

refinement to 400 outcomes. Its findings are supporting pilot evidence, separate from the full grid comparison.

Observed-incidence application

We aggregated corrected county-year SUD counts supplied by Habib¹² and SEER populations aged 18–64 into 58 community-college service-area clusters partitioning North Carolina's 100 counties. The corrected numerator includes heart-failure deaths previously excluded as presumed non-sudden (Habib, personal communication, September 25, 2026): 23,523 deaths over 25,594,321 person-years during 2018–2021. Annual rates were deaths/population per 100,000. Every county mapped to one cluster; annual cluster totals reproduced statewide totals. Observed rates were held fixed with no synthetic incidence generation. The four years were separate planning inputs, not longitudinal intervention follow-up.

Ethics and data use: [AUTHOR-SUPPLIED STATEMENT PENDING: responsible IRB/institution, approval or waiver/exemption determination and protocol number as applicable; consent determination and data-use permissions for restricted death-certificate sources and aggregate publication]. No approval or exemption is asserted by this placeholder.

Primary education outcomes used Equation 1 with $\beta = 0$, $\tau = 1$ and the named cached 58-cluster W . Incidence directed allocation only. The fit used intercept, treatment and true spillover without incidence adjustment. Thus the application does not assume historical SUD incidence predicts education performance or modifies the constant intervention effect.

All nine strategies were adapted before performance evaluation. Checkerboard became a deterministic greedy graph-interspersion assignment, not a literal checkerboard. Saturation used four frozen geographic regions selected from 250 k -means candidates by compactness and population/cluster/county balance, using pooled person-years/4 as reference population. Guided saturation ranked annual per-capita cluster rates, averaged cluster ranks equally within regions, then ordered regional saturations 80/60/40/20%. The grid rule averages its incidence covariate directly. The application mean-rank rule aligns with rank-scaled Poisson surfaces but differs from continuous grid incidence; it also differs from pooling deaths/population. Spatial blocking used 14 frozen yearly k -means groups based on standardized location, incidence ranks and log-population. Other strategies retained their mechanisms. SRS and incidence-balancing/targeting treated 29 clusters; Balanced Halves randomly assigned 14/15 within its two halves. Rounded local budgets were retained and reported rather than equalized retrospectively.

Primary queen settings crossed $\rho = 0, 0.2, 0.5$, $\gamma = 0.5, 0.8$, both regimes, four years and nine designs. Sensitivities added a hypothetical rank-scaled incidence baseline together with incidence adjustment, used rook parameter corners, or replaced guided saturation's regional mean rank with mean raw rates or pooled regional rates. The total was 1,248 reporting rows from 936 computational sources. Identical supports reused computation with explicit source IDs; annual budgets and population shares were

recalculated. Shared annual sources are not independent confirmations.

We drew 100 assignments and 100 independent outcomes per unique assignment, using 1,000 outcomes for proven singleton supports. Prespecified adaptive replication targeted relative MCSE(mean MSE) $\leq 5\%$ and MCSE(coverage) ≤ 0.01 ; duplicate-frequency covariance and both allocation/outcome variation were accounted for. A separate queen-corner extension retained assignments and raised stochastic outcome replication to 400. Opposite outcome halves selected and evaluated worst-decile assignments. Key-based seeds and manifest checks protected reproducibility and refused stale checkpoints. Further details, matched sensitivity tables and failure diagnostics are in the supplement.

Results

Grid accuracy and pilot downside

Table 1 reports the full queen benchmark at $\tau = 1$; Figure 1 illustrates regime differences in the detailed six-design subset. Guided saturation's pooled mean MSE was 0.0907 against SRS 0.1285, but regime separation was essential: control-only MSE was 0.1355 against 0.2146 (ratio 0.631), whereas both-arms MSE was 0.0459 against 0.0424 (ratio 1.083). Plain saturation had similar accuracy; an extra benefit from incidence guidance was not established. SRS and incidence balancing were competitive under both arms. Identified designs' coverage was approximately 94%, below 95%; rook Checkerboard was nonidentified and substantially biased. Rook results and effect-size/configuration breakdowns are reported separately in the supplement.

The selected-surface grid pilot supported saturation's control-only downside pattern. Guided saturation's estimated worst-decile conditional MSE was 0.1936 against SRS 0.3486; under both arms it was 0.0582 against 0.0544. Outcome refinement preserved the control-only direction but did not remove uncertainty about allocation tails. These are pilot values and do not replace full-study estimates.

Completed application

All 1,248 application reporting settings completed and passed precision gates. Production used 7,598,000 independent outcome fits; the separate extension added 2,280,000, for 9,878,000 distinct outcomes after counting copied prefixes once. Maximum relative MCSE(mean MSE) was 0.04871 and maximum MCSE(coverage) 0.00892. There were no warnings, aliases, boundary fits or estimation failures. All main stochastic sources passed at 100 assignments and 100 outcomes; two singleton baseline-sensitivity sources required 2,000 outcomes.

Under control-only spillover, plain saturation's yearly descriptive mean MSE was 0.22329 against SRS 0.30761, a 27.4% reduction. Guided saturation's reductions were 21.1–29.9% depending on year (Table 2, Figure 2). Both saturation designs had lower MSE than SRS in every primary parameter setting, including setting-specific approximate MC intervals. Neither was uniquely best. High Incidence Focus's control-only accuracy varied markedly across years.

Under both arms, SRS mean MSE was 0.07638, plain saturation 0.07518 and guided saturation 0.07428–0.07682. These small differences do not establish equivalence or universal optimality. Primary yearly mean coverage was approximately 92.6–93.6%, below nominal; passing precision targets did not repair undercoverage.

At the refined queen corners with $\gamma = 0.8$, control-only plain saturation had estimated q90 31.8–32.9% below SRS and worst-decile mean 34.6–37.8% below; guided saturation reductions were 20.8–38.6% and 20.7–43.4%. Cross-selected outcome halves retained this pattern (Figure 3). However, only ten sampled draws formed the worst decile. All 112 non-singleton reporting rows flagged uncertain tail-boundary membership. Both-arms guided saturation had half-sample Jaccard overlap 0–0.25, with about 61–86% of raw estimated-risk variance attributable to outcome MC noise. Small both-arms tail differences cannot support superior downside protection. Singleton Checkerboard and High Incidence Focus have no allocation variation for these inputs, but can still have substantial estimation error.

Budgets differed (Figure 4). Graph Checkerboard treated 27 clusters under queen; buffer averaged 15.89; spatial blocking treated 28/31/27/29 by year, including a singleton group always assigned control under the retained rounding rule. High Incidence Focus treated 29 clusters but approximately 22–24% of working-age population; buffer reached about 27%. These shares describe geographic reach, not healthcare-professional enrollment or cost. Graph Checkerboard was identified but imprecise, unlike the exact rectangular-rook alias.

The hypothetical baseline/X-adjusted sensitivity increased High Incidence Focus MSE roughly 3.68–4.44-fold under both arms and 1.42–2.52-fold control-only. Treatment-incidence alignment reduced information after adjustment; baseline and adjustment changed jointly. Matched rook/queen corners preserved broad conclusions. Alternative regional summaries changed some region orders, with no retrospective replacement of the prespecified mean-rank rule.

Discussion

The completed grid and observed-incidence application support conditional guidance: consider saturation when spillover primarily reaches controls, and retain SRS among competitive options when both arms receive exposure. Average MSE and allocation downside should be evaluated together. A restricted design's near-zero allocation variability is insufficient if its conditional error is high. Conversely, small estimated tail differences with noisy membership are insufficient to dismiss SRS.

Incidence guidance did not establish an accuracy advantage over plain saturation. Historical incidence can still express implementation priorities, but that purpose should be distinguished from improved estimation. High Incidence Focus illustrates the trade-off: low conditional both-arms error in the primary education model came from a fixed assignment, not a demonstration of randomized causal identification. Its sensitivity to incidence adjustment also prevents a general recommendation based on that result alone.

Table 1. Full grid benchmark at $\tau = 1$, queen contiguity. Each regime-specific entry averages 80 scenarios (five incidence configurations, four ρ and four γ values). MSE and coverage use the oracle fit. SRS is complete randomization of 50 clusters. No value pools queen and rook.

Design	Both arms		Control only	
	MSE	Coverage	MSE	Coverage
Checkerboard	0.0624	0.940	2.1114	0.942
High Incidence Focus	0.1617	0.941	0.3292	0.935
Saturation Quadrants	0.0442	0.942	0.1359	0.929
Isolation Buffer	0.1608	0.939	0.1595	0.939
2×2 Blocking	0.0623	0.937	0.5761	0.941
Balanced Quartiles	0.0421	0.940	0.2195	0.932
Balanced Halves	0.0420	0.939	0.2179	0.933
Incidence-Guided Saturation	0.0459	0.939	0.1355	0.934
SRS	0.0424	0.941	0.2146	0.936

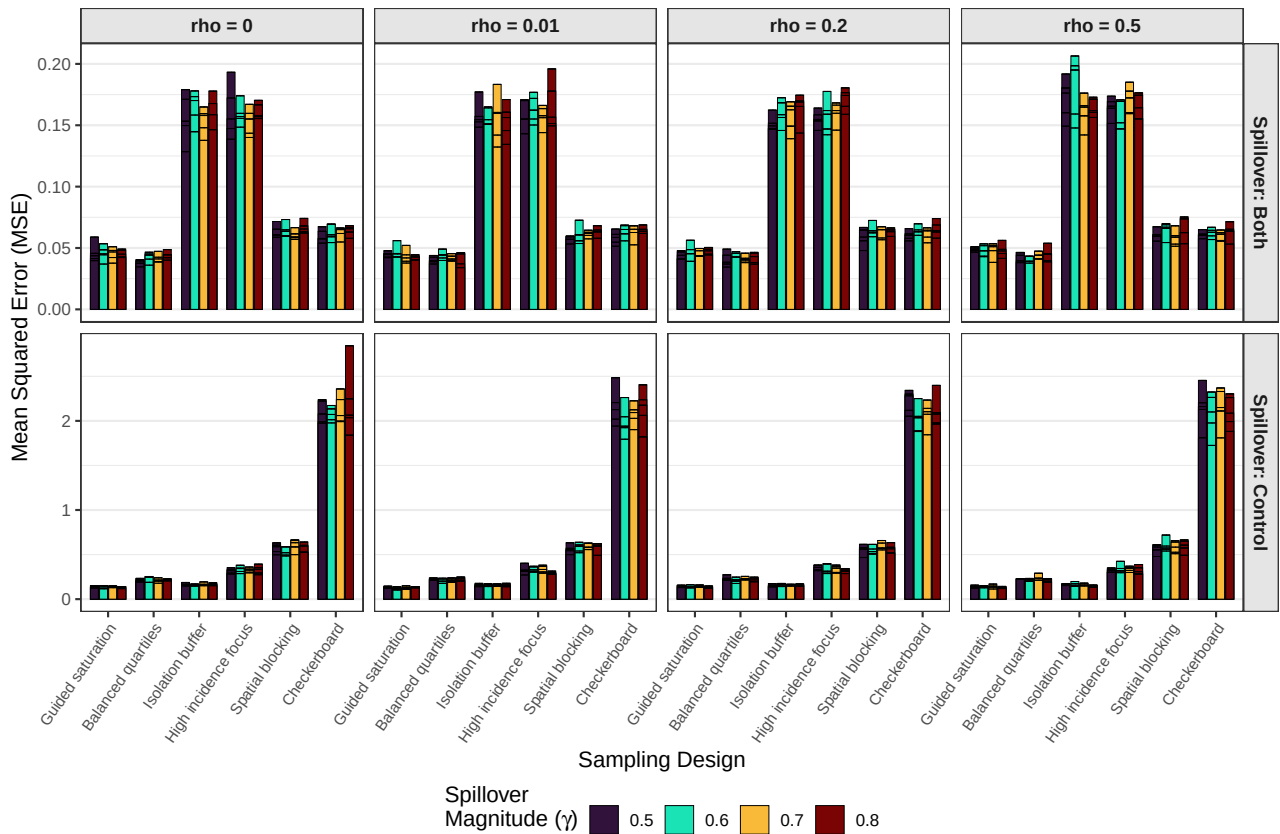


Figure 1. Grid MSE at $\tau = 1$ under queen contiguity for the detailed six-design subset. Columns show outcome spatial dependence and rows spillover regimes. Bars overlay five incidence configurations, with horizontal lines marking their values; colors show spillover magnitude. Table 1 additionally includes plain saturation, Balanced Halves and SRS. Regime scales differ.

Table 2. Primary application yearly mean MSE, queen contiguity; equal weights over the six ρ/γ pairs. Plain saturation and SRS repeat shared primary sources across years.

Year	Both arms		SRS	Control only	
	SRS	Guided sat.		Plain sat.	Guided sat.
2018	0.07638	0.07428	0.30761	0.22329	0.24285
2019	0.07638	0.07628	0.30761	0.22329	0.21565
2020	0.07638	0.07682	0.30761	0.22329	0.22029
2021	0.07638	0.07450	0.30761	0.22329	0.23732

These comparisons complement spatial analysis rather than replace it. The primary fit knows the spillover mechanism; the grid sensitivity omitting it showed bias and poor coverage. We did not establish performance with incorrect exposure networks, heterogeneous treatment

effects or participant-level sampling error. The grid and one irregular geography do not span all trial structures, and unequal treatment budgets prevent interpreting the rankings as equal-cost optimization. Historical incidence was fixed;

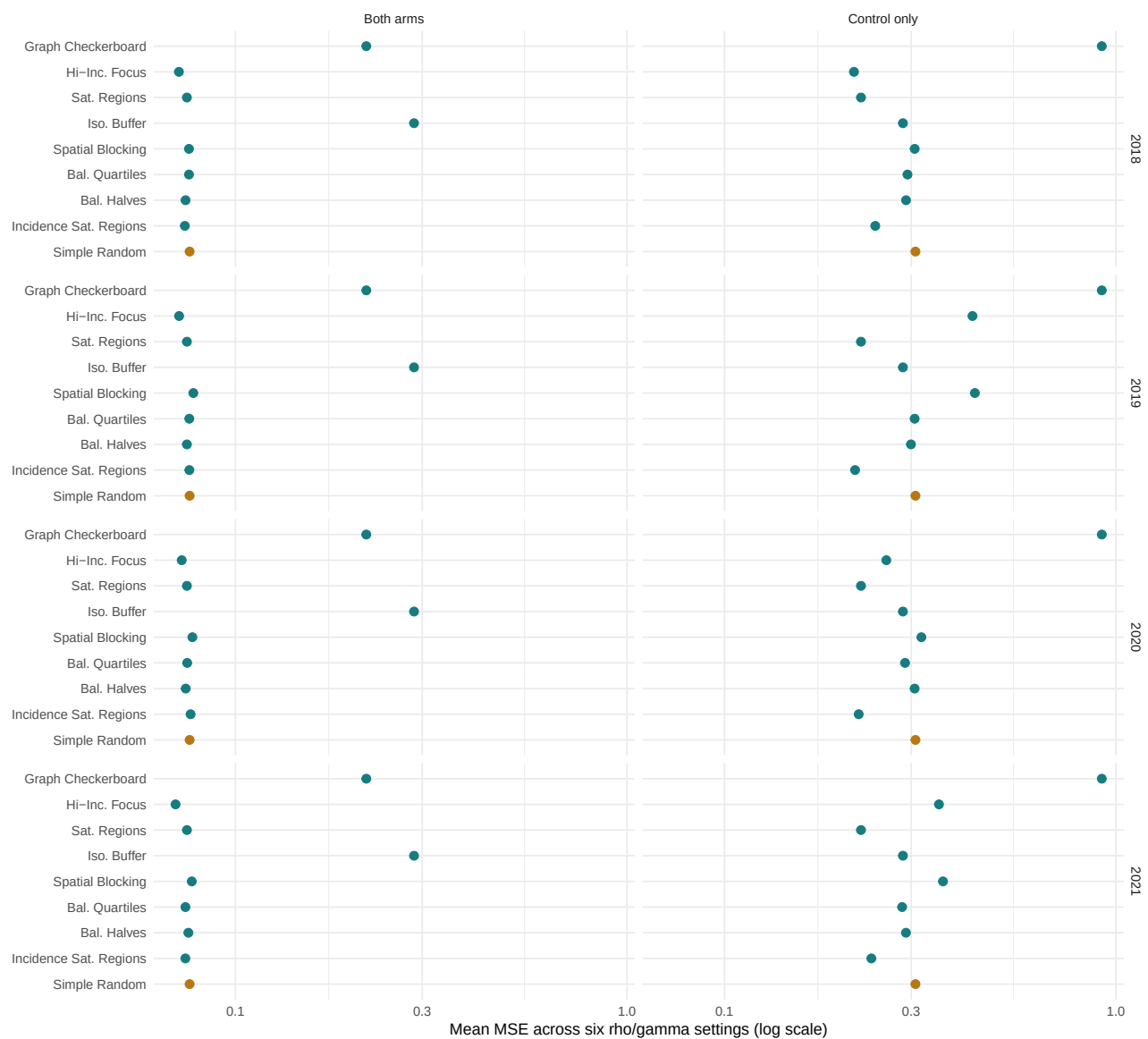


Figure 2. Completed primary application mean MSE by design, observed planning year and spillover regime, queen weights and education outcome. Each point equally averages six ρ/γ pairs on a common logarithmic scale; gold marks SRS. Shared sources repeat across years and are not independent replications. No pooled MC intervals are supplied.

measurement uncertainty and change between planning and implementation were not propagated.

Wald coverage remained below nominal even when Monte Carlo precision was adequate. Interval calibration must be addressed before adopting this analysis for a final protocol. Allocation-tail estimates are conditional on sampled masks and finite outcome replication; the extension improved risk estimation for the same masks rather than providing fresh allocation confirmation. Shared annual supports also preclude treating repeated performance as independent evidence.

The education application is a completed design evaluation, not an implemented intervention or evidence that education reduces SUD. Final decisions require a specified education endpoint, eligible professionals, budget and plausible communication network. Within the evaluated assumptions, saturation offers favorable average and descriptive downside performance under control-only

spillover; SRS remains a reasonable comparator and candidate under both-arms exposure.

Acknowledgements

OpenAI Codex assisted with code implementation and checking, data visualization and manuscript drafting. Numerical results were generated by the documented R simulations and maximum likelihood fits. The authors retain responsibility for verification, interpretation and the final submitted text.

Declaration of conflicting interests

The authors declare that there is no conflict of interest.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.



Figure 3. Refined allocation-risk estimates at primary queen $\rho = 0.5, \gamma = 0.8$. Mean MSE, estimated allocation q90 and estimated worst-decile mean refer to the same 100 sampled assignments, using 400 outcomes per stochastic assignment or 1,000 for singleton supports. Finite-outcome tails and membership remain uncertain; this extension is not fresh allocation confirmation.

Data availability

Simulation code, manifests and aggregate outputs are maintained in the SpatialCRT repository (<https://github.com/andrew-walther/SpatialCRT>), under projects/IncidenceDesign. Restricted county death-certificate source files and individual-fit caches are excluded. Authorized 58-cluster summaries and reproducible aggregate results accompany the application; access to restricted sources is subject to institutional permissions. The submission checklist records the remaining ethics/data-use documentation.

additionally displays application development, models and filterable findings.

Supplementary material

The supplementary document supplies exact design rules, parameter grids, estimation diagnostics, detailed grid results, application maps, annual performance/budgets and matched sensitivities. The offline HTML companion

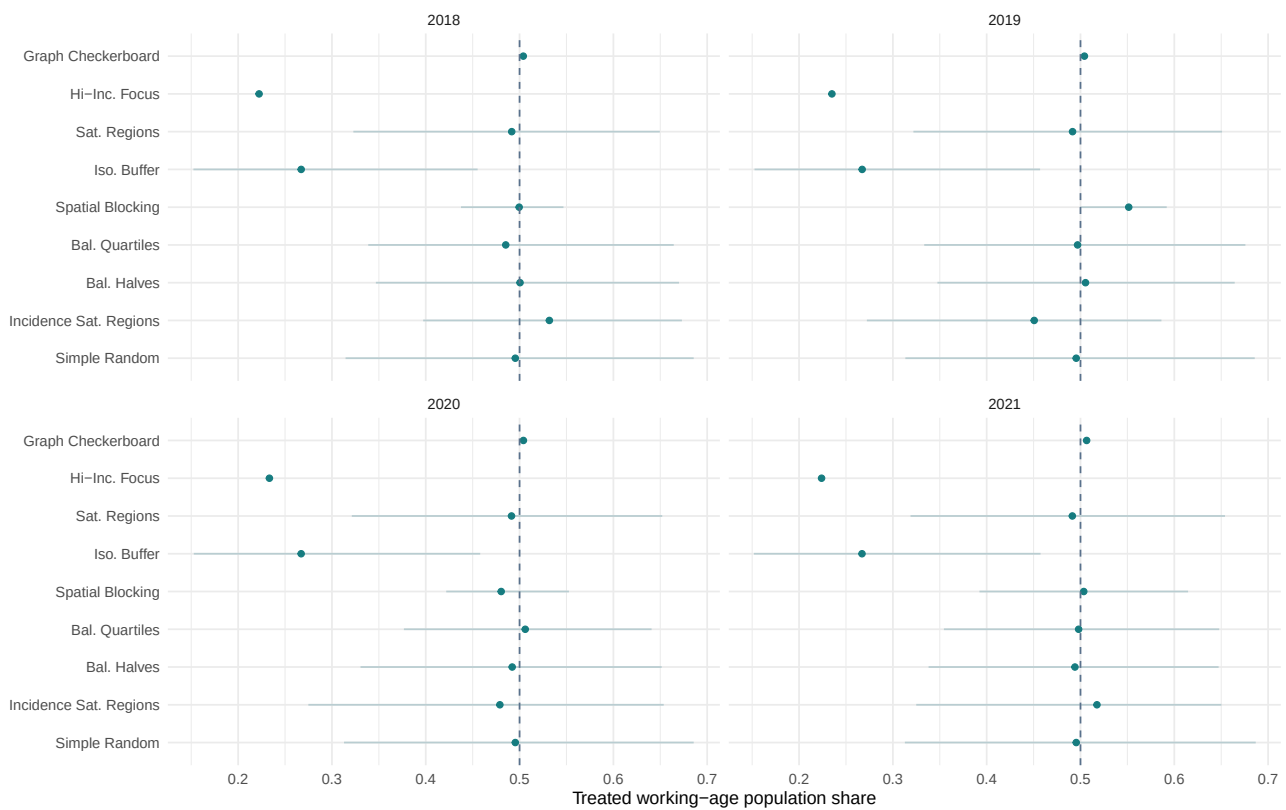


Figure 4. Treated annual working-age population shares under primary queen assignments. Points are means and segments show sampled allocation ranges; the dashed line marks 50%. Budgets use the both-arms run, with allocation support unchanged by regime. Population is geographic reach, not healthcare-professional enrollment or implementation cost.

References

1. Donner A and Klar N. *Design and analysis of cluster randomization trials in health research*. Arnold, 2000. ISBN 978-0-470-71100-2 978-0-340-69153-3.
2. Hayes RJ and Moulton LH. *Cluster Randomised Trials, Second Edition*. Chapman and Hall/CRC, 2017. ISBN 978-1-315-37028-6. DOI:10.4324/9781315370286. URL <https://www.taylorfrancis.com/books/9781315370286>.
3. Hudgens MG and Halloran ME. Toward Causal Inference With Interference. *Journal of the American Statistical Association* 2008; 103(482): 832–842. DOI:10.1198/016214508000000292. URL <https://www.tandfonline.com/doi/full/10.1198/016214508000000292>.
4. Jarvis C, Di Tanna GL, Lewis D et al. Spatial analysis of cluster randomised trials: a systematic review of analysis methods. *Emerging Themes in Epidemiology* 2017; 14(1): 12. DOI:10.1186/s12982-017-0066-2. URL <https://ete-online.biomedcentral.com/articles/10.1186/s12982-017-0066-2>.
5. Crisp AM, Halloran ME, Longini IM et al. Covariate-constrained randomization with cluster selection and substitution. *Clinical Trials* 2023; 20(3): 284–292. DOI:10.1177/17407745231160556. URL <https://journals.sagepub.com/doi/10.1177/17407745231160556>.
6. Baird S, Bohren JA, McIntosh C et al. Optimal design of experiments in the presence of interference. *The Review of Economics and Statistics* 2018; 100(5): 844–860. DOI:10.1162/rest_a.00716.
7. Cai C, Zhang X and Airolidi EM. Independent-set design of experiments for estimating treatment and spillover effects under network interference, 2023. 2312.04026.
8. Leung MP. Rate-optimal cluster-randomized designs for spatial interference. *The Annals of Statistics* 2022; 50(5): 3064–3087. DOI:10.1214/22-AOS2224.
9. Walther A, Van Deirse T and Lin FC. Spatial cluster randomized trials - sampling design with spillover effects & spatial dependence. *BMC Medical Research Methodology* 2026; DOI:10.1186/s12874-026-02968-0. URL <https://link.springer.com/article/10.1186/s12874-026-02968-0>. Published online 26 August 2026.
10. Ord K. Estimation Methods for Models of Spatial Interaction. *Journal of the American Statistical Association* 1975; 70(349): 120–126. DOI:10.1080/01621459.1975.10480272. URL <http://www.tandfonline.com/doi/abs/10.1080/01621459.1975.10480272>.
11. Bivand R and Piras G. Comparing Implementations of Estimation Methods for Spatial Econometrics. *Journal of Statistical Software* 2015; 63(18). DOI:10.18637/jss.v063.i18. URL <http://www.jstatsoft.org/v63/i18/>.
12. Habib A. *Temporal Trends and Social Determinants of Sudden Death Mortality in North Carolina*. Master's paper, Department of Biostatistics, University of North Carolina at Chapel Hill, Chapel Hill, NC, 2026. Unpublished.